# 📽️ SmartTrace - Project Presentation

**Team Name**: Runtime Reepers

**Theme**: Supply Chain Transparency / Anti-Counterfeit

# SmartTrace

## The Living Supply Chain Ledger

"Immutable Authenticity. From Factory Floor to Patient Hands."

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"Good afternoon. We are Runtime Reepers, and this is SmartTrace. We are tackling a **$200 billion global crisis—counterfeit pharmaceuticals**—by building a digital thread of authenticity that connects every single pill to its source."

# The Problem: The "Black Box" of Logistics

### The Gap

Once products leave the manufacturer, visibility is lost until delivery.

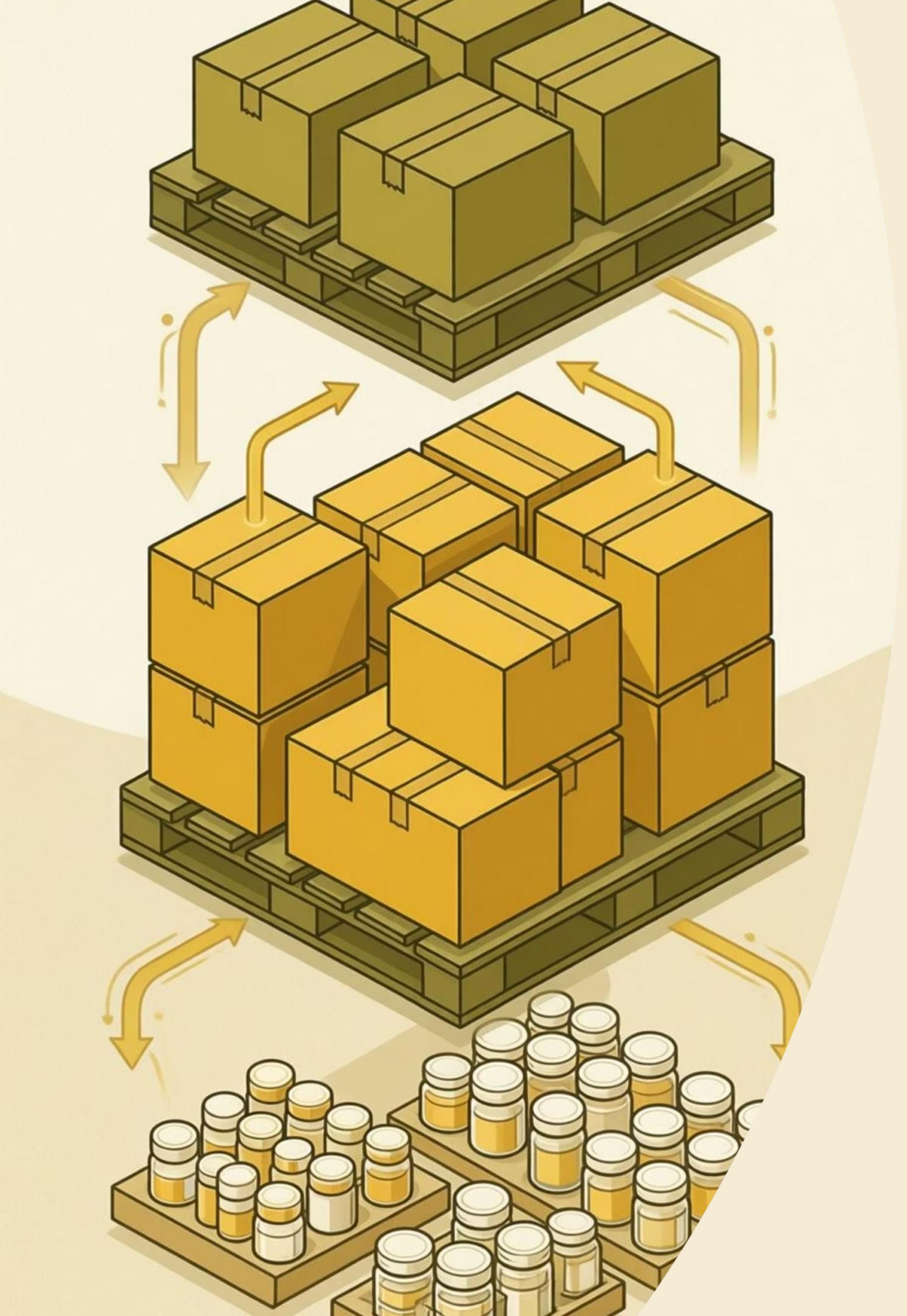
### The Risk

Diverted cargo, counterfeit swaps, and unsafe storage conditions.

### The Cost

Brand reputation loss and, most importantly, patient safety.

"Manufacturers know what they make, and pharmacies know what they buy. But the middle mile—the logistics network—is a black box. This is where verified drugs are swapped for fakes or diverted to gray markets. We set out to illuminate this black box."



# The Solution: Granular Traceability

### Hierarchy-Aware

We don't just track shipments; we track **Pallets ↣ Cartons ↣ Units**.

### Inheritance

If a Pallet is flagged, every individual Unit inside it is instantly marked "Suspicious" worldwide.

### Digital Thread

Every item has a unique digital twin anchored in our ledger.

"SmartTrace isn't just a database; it represents the real world. We treat items as nested assets. If a pallet is compromised, our system understands that the 500 units inside it are also compromised. We link them cryptographically, creating a chain of custody that cannot be broken."

# Core Tech: Geo-Fencing & Anomaly Detection

## Smart Location Logic

- Compares Expected Location (Distributor's Route) vs. Actual Location (Scan Coordinates).
- **Logic**: If Distance(Expected, Actual) > Threshold THEN Status = 'SUSPICIOUS'.

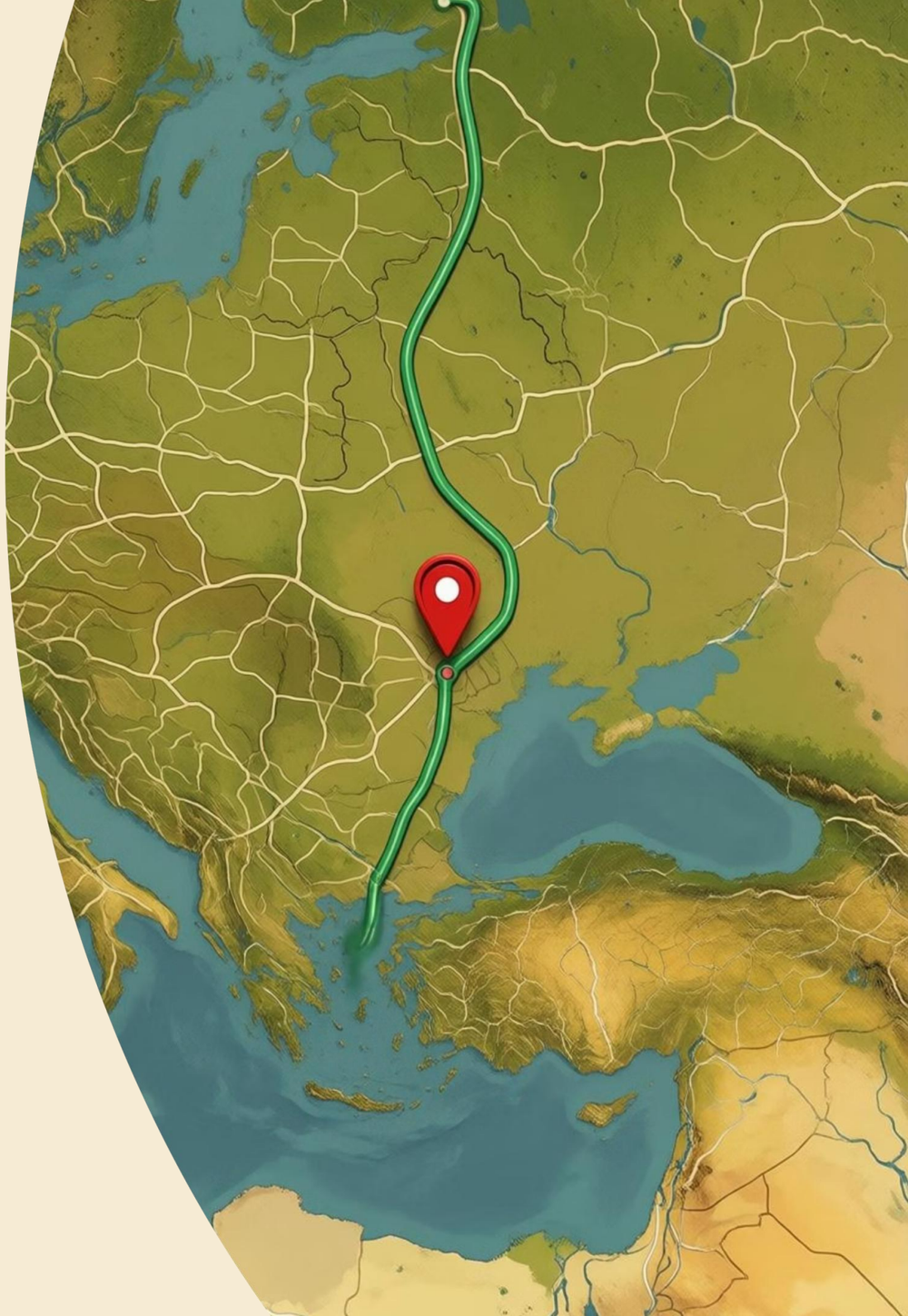
### ⚠️ Anomaly Detected

Scan distance > 5km from warehouse

### Real-Time Intervention

Manufacturers can freeze a batch instantly if diversion is detected.

"This isn't just a passive record; it's an active guardian. Our Anomaly Detection Engine calculates the geodesic distance between where a product *should* be and where it was *scanned*. If a Mumbai distributor's package appears in Delhi, the system instantly flags it, preventing diversion in real-time."



# User Experience: Glassmorphic & Seamless

## Design First

Implemented with a "State-of-the-Art" Glassmorphism UI to inspire trust.

## Accessibility

- **Staff Portal**: For manufacturers to manage inventory and view alerts.
- **Public Verify**: No login required. Anyone can scan to verify authenticity.
- **Tech**: Powered by framer-motion for fluid 60fps animations.

"Security doesn't have to be ugly. We built SmartTrace with a consumer-grade user experience using modern glassmorphism trends. It's fast, responsive, and intuitive—whether you're a warehouse logistics manager or a concerned parent scanning medicine at home."

# The Technology Stack

### Frontend

Next.js 16 (App Router) & React 19

### Styling

TailwindCSS v4 & Framer Motion

### Database

MongoDB (Mongoose) - *For complex polymorphic data modeling*

### Auth

NextAuth.js - *Role-Based Access Control (RBAC)*

### Validation

Zod & TypeScript - *Strict type safety for critical supply chain data*

"We built this on the modern web's most robust stack. We use Next.js 16 for server-side performance, MongoDB to handle the complex nested relationships of our supply chain items, and Zod to ensure that absolutely no bad data enters our trusted ledger."

# Future Roadmap

### Blockchain Integration

Porting the Ledger to **Solana/Ethereum** for decentralized immutability.

### IoT Integration

Passive RFID scanning at warehouse doors (no manual scanning needed).

### AI Predictions

Predicting supply shortages based on scan velocity in different regions.

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# "SmartTrace: Trust, Verified."

"We've built the foundation of truth. Our next step is to make it decentralized and autonomous. SmartTrace is more than an app; it's the future standard for integrity in global trade. Thank you."